

Midterm Project 1

This is an *individual* effort. You can come to me or the TAs for help if you get stuck, but you should not be working in groups. Class notes and anything linked from the class ELMS page can be used as a resource. Submit both your RScript code and the requested word document with your name and the assignment in the file name.

Section 1: Binomial Data

Description:

Cancer in dogs: A study in 1994 examined 491 dogs that had developed cancer and 945 dogs as a control group to determine whether there is an increased risk of cancer in dogs that are exposed to the herbicide 2,4-Dichlorophenoxyacetic acid (2,4-D). Order: a factor with levels 2,4-D and no 2,4-D, response: a factor with levels cancer and no cancer. This was an observational study

Questions: Write your answers in a word file to be submitted with your RScript file.

1. Identify the control group and the treatment group.

The control group in this study is the dogs that were not exposed to 2,4-D. The treatment group is the dogs that were exposed to 2,4-D.

2. Write a hypothesis for what you think the effect of exposure to the herbicide might be as shown in class both in words and mathematically.

The effect of exposure to the herbicide 2,4-Dichlorophenoxyacetic acid (2,4-D) will increase the risk of developing cancer in dogs. Specifically, it is expected that dogs exposed to 2,4-D will have a higher chance of getting cancer compared to dogs not exposed to this herbicide.

Math terms:

Null hypothesis H_0 : There is no significant difference in the risk of cancer between dogs exposed to 2,4-D and dogs not exposed to 2,4-D.

Alternative hypothesis H_a : There is a significant difference in the risk of cancer between dogs exposed to 2,4-D and dogs not exposed to 2,4-D.

3. Now test that hypothesis. Select an alpha, choose the correct binomial hypothesis test and run it.

Rstudio

4. What are your results? Express these as shown in class.

Results from R: My p-value was very close to 1 and my alpha value was .05. In this case, we fail to reject the null hypothesis. There is no significant difference in the risk of cancer between dogs exposed to 2,4-D and dogs not exposed to 2,4-D.

5. Does this study make it possible to prove causation? Why or why not?

A p-value very close to 1 suggests that there is no statistically significant evidence to reject the null hypothesis. That being said, we cannot prove causation. The study's lack of statistical significance does not directly address the issue of causation. Additional research and evidence from various sources would be needed to establish a causal relationship between 2,4-D exposure and the risk of cancer in dogs.

Section 2: Test of Means

This data approaches student achievement in secondary education of two Portuguese schools. The data attributes include student grades, demographic, social and school related features, and it was collected by using school reports and questionnaires. Two datasets are provided for two distinct subjects: Mathematics and Portuguese.

1. Compare the mean G3 grades (the equivalent of the semester grade) for Mathematics and Portuguese.

1. What test are you going to run?

I am going to run a two-sample unpaired t-test since we are comparing the means of two independent groups, in this case, the mean G3 grades for Mathematics and Portuguese subjects. The t-test will help determine if there is a statistically significant difference between the mean grades of the two subjects.

2. Have you met all assumptions?

Yes, I've checked for normality, independence and, equal variances, check RStudio.

3. What are your null and research hypotheses and your alpha in words and in mathematical notation?

Null Hypothesis (H0): The mean G3 grade for Mathematics is equal to the mean G3 grade for Portuguese.

Alternative Hypothesis (Ha): The mean G3 grade for Mathematics is not equal to the mean G3 grade for Portuguese.

Significance level (Alpha value) is .05.

4. What are your results?

Two Sample t-test

```
data: mathG3 and porG3
t = -6.1504, df = 1042, p-value = 1.1e-09
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.966454 -1.015184
sample estimates:
mean of x mean of y
 10.41519  11.90601
```

The results of the paired t-test suggest that there is a statistically significant difference between the mean math grades (mathG3) and the mean Portuguese grades (porG3). The t-value is -6.1504, and the p-value is very low (1.1e-09, which is close to 0). The t-value indicates the size of the difference between the means relative to the variability in the data, while the p-value indicates the probability of observing such a difference if the null hypothesis (no difference in means) were true.

In this case, the very low p-value (below 0.05) suggests strong evidence against the null hypothesis.

We reject the null hypothesis, the mean G3 grade for Mathematics is not equal to the mean G3 grade for Portuguese since the 95% confidence interval for the difference in means is from -1.966454 to -1.015184 and the difference in means is greater than these values.

2. Compare the G1 and G2 mean grades for either Mathematics or for Portuguese (just one of the datasets). Each row of data represents information for a single student.

1. What test are you going to run?

To compare the mean grades between G1 and G2 for Mathematics I will use a paired t-test. This is an appropriate test since I have two related groups (in this case, G1 and G2 grades for the same students) and I want to determine if there is a significant difference in the means of paired observations.

2. Have you met all assumptions?

Yes, I've checked for normality, independence and, equal variances, check RStudio.

3. What are your null and research hypotheses and your alpha in words and in mathematical notation?

Null Hypothesis (H0): The mean G1 grade is equal to the mean G2 grade for either Mathematics.

Alternative Hypothesis (Ha): The mean G1 grade is not equal to the mean G2 grade for either Mathematics or Portuguese

Select Significance Level (Alpha): .05

4. What are your results?

**Paired t-test data: mathG2 and mathG3 $t = 2.9869$, $df = 394$, $p\text{-value} = 0.002994$
alternative hypothesis: true mean difference is not equal to 0 95 percent
confidence interval: 0.1021055 0.4953629 sample estimates: mean difference
0.2987342. The p-value of 0.002994 is less than the common significance level of 0.05. This indicates strong evidence against the null hypothesis that the means of mathG2 and mathG3 are equal. The estimated mean difference between mathG2 and mathG3 is 0.2987342. This means that, on average, there is a difference of approximately 0.30 between the math grades in these two groups.**

Based on the t-test results and assuming a significance level of 0.05, I reject the null hypothesis and conclude that there is a significant difference between the mean math grades in groups mathG2 and mathG3. The means of these two groups are not the same; there is a statistically significant difference between them.

Section 3: Concepts

1. Which of the following IS/ARE NOT a measure of central tendency?

1. Sample -**is not**
2. Mean- **is**
3. Range- **is not**
4. Median- **is**
5. Mode- **is**

2. Define each of the following:

1. Probability distribution-

A probability distribution is a mathematical function that gives the likelihood of several probable outcomes for a given random variable.

2. Population

Population refers to the entire set of people, objects, or events that statistical researchers are interested in studying.

3. Sample

A sample is a portion of people, things, or events chosen from a larger population. Sampling is done to collect data from a manageable and representative subset of the population so that conclusions or inferences about the complete population may be drawn.

4. Parameter

A parameter is a measure or attribute used to characterize a population. Numerical numbers known as parameters serve to condense significant facets of the behavior or distribution of a population.

5. Variable

A variable is any feature or trait that may be assigned values or categories. In a research, experiment, or observation, variables are used to measure, describe, or categorize significant elements.

3. For each of the following, identify the variable type:

1. Dice rolls- **Discrete numerical variable**
2. Lyme disease infection rate- **continuous, numerical**
3. Satisfaction survey rankings- **ordinal, categorical**
4. Whether or not a dog got parvo- **Binary, categorical**
5. IQ scores- **Continuous, Numeric**
6. Ethnicity- **Categorical**
7. Height- **continuous, numerical**
8. Income- **continuous, numerical**

4. Describe three types of probability sampling and when you would use each.

1) Random sample: Every member of a population has an equal chance of being selected. Random sampling is used when the population is relatively small and homogeneous. It would be used when all elements in the population are similar in characteristics of interest.

2) Stratified random: split populations into groups and randomly select from each group. When a population is diversified and divided into discrete

subgroups with different traits, stratified random sampling might be helpful. By guaranteeing that every subgroup is included in the sample, it makes it possible to compare and analyze data both within and across strata.

3) Systematic random: Put every member of the population in some order. Choose a random starting point and select every n th member from that order or list. The sampling interval (n) is determined based on the desired sample size and the population size. When a population is big and well-organized, members of the population can be readily named or arranged in a certain order—systematic sampling is appropriate (a list of students at a school). In situations when the list is random or the order does not create bias, it is more effective and straightforward to implement than basic random sampling for large populations, while still yielding a representative sample.

5. If a test is listed as robust with respect to normality, what does that mean?

If a test is listed as robust with respect to normality, this means that even if the data slightly deviates from a normal distribution, the test may still provide accurate findings and valid statistical inference. An example of a robust test is an Anova test since the assumption of normality can be inaccurate and can still provide valid results.

6. Identify three ways in which statistics can be used incorrectly or misleadingly. Give an example of each. How would you respond if asked to do one of these things at a job?

1) Cherry-picking data: Selecting particular data points, time periods, or subsets of data to support a certain notion or argument while ignoring evidence that may contradict it or offer a more nuanced view is an unethical activity. If I were asked to cherry-pick data, I would instead try to report all data and report everything including any errors. To present a more favorable outcome, it's crucial to provide a comprehensive and balanced analysis that includes all relevant data points and time periods. Transparency and honesty in data reporting are essential.

2) P-hacking is the process of doing one or more statistical tests on a particular set of data until a statistically significant result (usually a p -value less than a certain threshold, such as 0.05) is produced. This may entail experimenting with various analytic techniques like eliminating outliers or changing variables to get the intended result. P-hacking inflates the possibility of discovering significant findings by chance alone, which can result in false-positive results and incorrect conclusions. If I were asked to use this practice, I would instead correct the p -value by using Bonferroni

correction. I would also decide my p-value before I run my tests to make sure that there are statistically significant findings.

3) **HARKING** is the process of formulating research questions or hypotheses after data analysis and outcomes are obtained, as opposed to doing so before data collection and analysis. Post-hoc analysis techniques like **HARKING** might result in incorrect conclusions or misinterpreted data if they are not properly taken into consideration. If I were asked to practice this technique at a job, I would report that it is a post-hoc analysis and try to clearly define my hypotheses or research questions before collecting data and analysis.

7. Why is a large sample size an advantage? What problems does it solve?
Why do researchers not always take large samples?

An advantage of having a large sample size is that it yields a more representative sample of the population. It makes the results more broadly applicable and ensures that the sample accurately reflects the variation and diversity seen in the population. A large sample size is also useful when trying to prove the precision and accuracy of the statistical study. Population parameter estimates, including means, proportions, or correlations, are often more accurate and exact. Larger samples provide more consistent results by reducing the effects of sampling error and random variation. Large samples may be expensive, labor-intensive, and time-consuming, which is why researchers may not always enjoy them. Particularly in large-scale studies or longitudinal research, researchers may encounter financial restrictions, or logistical difficulties. With bigger samples, it becomes more difficult to ensure data accuracy, and validity.

8. Under what (if any) circumstances can causation be proved?

There are not many circumstances where causation can be proved. For example, if two variables are correlated, we will see a pattern when both variables are plotted on a scatter plot. Although we may see a pattern, we cannot prove that one variable causes a change in the other. The relationship between both variables could be coincidental, or there may be a third factor that is causing both variables to change.

9. I have two normally distributed samples from two different populations. I do not know if the populations have the same variance. What test would I use to compare the means of those samples and why?

You should use Welch's T-test since it doesn't assume equal variances between the populations. For this situation, since we don't know if the

populations have the same variance, we can establish an accurate result using Welch's T-test.

10. What happens if you violate the assumptions of a test?

If you violate the assumptions of a test, then the results drawn from the analysis can be invalid and may not accurately reflect the underlying relationships or effects in the data. Predictive models of the data can also be inaccurate if you violate the assumptions of the test.

11. Can a confidence interval take the place of a t-test? Why or why not?

No, a confidence interval cannot take the place of a t-test. Although they are connected to estimating and comparing means, t-tests and confidence intervals have different uses. A confidence interval indicates the range of plausible values for a population parameter (such as a mean), but it does not explicitly test whether that parameter differs significantly from another value. A t-test is a hypothesis test that determines the statistical significance of a difference in means between groups or between a sample mean and the predicted population mean. Using a confidence interval would not result in a p-value or a test statistic to determine the probability that the observed difference is the result of chance. It wouldn't have the foundation for hypothesis testing that is required to determine the statistical significance of differences.